# Self-Supervised Machine Learning on MRI Data

## Interim report

by

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# 1 Introduction

Lesion segmentation plays a critical role in a variety clinical tasks such as medical diagnosis, prognosis, and treatment [1]. In most cases, it is performed manually by the radiologist, making the process very time-consuming and dependent on understanding of the human anatomy and expertise. In addition, human predictions can be skewed by factors such as inattentive blindness, which can lead to high miss-rates of unexpected anomalies [2].

Recent advances in deep learning, and generative modeling in particular, have proven to be promising tools for detecting arbitrary anomalies in data and automating medical analysis, relieving professionals from time-consuming tasks [3]. However, training supervised models can often be a significant challenge. The reason for this is that training neural networks typically requires large amounts of labeled data, which is often difficult to obtain due to disease heterogeneity and patient privacy aspects. Therefore, we need to consider such machine learning models that do not require vast amounts of manually labeled data and can properly generalize to rare diseases that may not have been seen during training.

One of the possible approaches is to use machine learning models based on Variational Inference (See Section 5.2.2) to model the healthy distribution of the human brain in a low-dimensional space and then, during inference, detect anomalous data passed as input to that model as outlier. This approach is more similar to how radiologists make their decisions, as it does not require manual annotation and relies only on a priori knowledge of pathology manifestations. [4].

Although there are several possible ways to address this task, this project will focus mainly on *probabilistic graphical models* such as Variational Autoencoder (VAE) [5] and it's extensions [6,7]. VAEs learn the distribution of the data by mapping it to the normative distribution in the *latent space* and do not require any labels for learning. Once trained, the model will produce different latent encodings for the unhealthy samples as they will have a different distribution than the one it has learned. We can then try to segment the anomalous area, either by applying post-processing techniques to the reconstructions or by analyzing the latent space.

# 2 Objectives

In this project, I will focus on implementing and evaluating some of the existing solutions to the brain tumor segmentation problem, and exploring how well models can adapt to data coming from new domains. In addition, I will discuss how the difference in the latent space size affects the quality of segmentation and what are some of the weaknesses of the described methods. The specific objectives of this study can be formulated as follows:

### 2.1 Primary Objectives

1. Investigate and compare some of the available medical image analysis methods for self-supervised lesion detection and segmentation.
2. Implement the described architectures in Python using PyTorch deep learning framework.
3. Examine the performance of Autoencoders (AE) and Variational Autoencoders in relation to lesion detection and segmentation tasks.
4. Research the effects of domain shift and the ability of models to generalize to new data with different distributions.

### 2.2 Extension Objectives

1. Explore the extensions of VAEs and compare their results to basic VAE and AE results.
2. Study the weaknesses of the demonstrated methods and discuss potential improvements.
3. Discuss if the amount of training data used is sufficient for proper model generalization and inference.
4. Study how the size of the latent space affects the quality of segmentation.

## 3 Related Work

### 3.1 Unsupervised Anomaly Detection

Unsupervised anomaly detection (UAD) is of interest in the field of medical imaging due to its potential to detect and localize malignant structures in the absence of massive annotated training datasets. Traditional approaches to anomaly detection have been based on statistical methods, both *parametric* (Gaussian mixture modeling, hidden Markov models) and *non-parametric* (histogram analysis, k-nearest neighbors) [8]. Deep learning emerged as a promising tool for detecting anomalies in medical data, largely due to generative-based methods, which will be explored in this study.

The two main groups of generative methods are based on Generative Adversarial Networks (GANs) [9] and Variational Bayesian models [10]. GANs can learn to generate realistic high-dimensional images similar to the ones seen during training by sampling from a lower-dimensional latent space, while variational methods such as VAE are used to map data onto its latent representation and then sample from the learned distribution.

However, volumetric data modeling remains a challenge due to the number of parameters to be learned and the lack of computational resources. [11] overcomes this by training VAE on 2D slices and then combining it with a Gaussian model that captures relationships between those slices. In this approach, the latent space mean and variance are estimated for each slice dimension over many different volumes, producing a distribution of 3D data from which we can sample. The general process of brain anomaly segmentation is shown in Fig. 1.

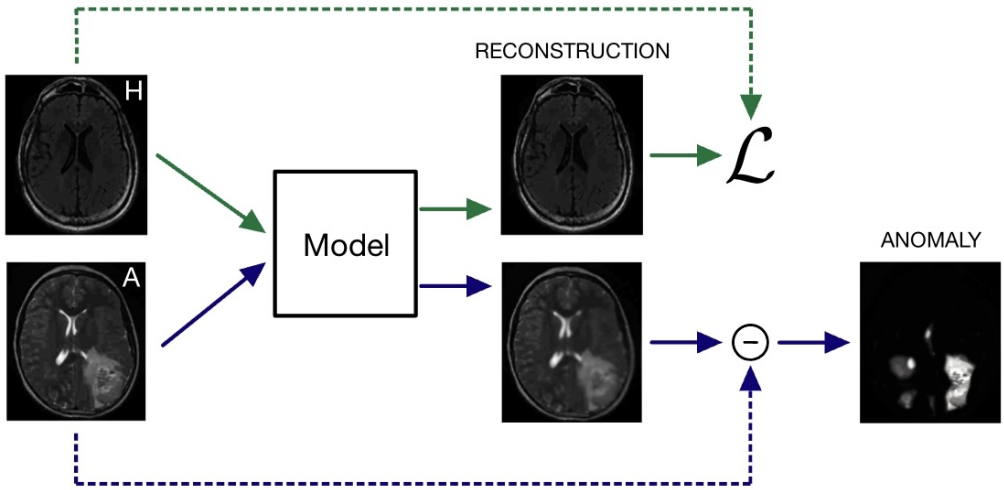


Figure 1: Unsupervised anomaly detection. **H** corresponds to a healthy brain image and **A** corresponds to an abnormal one. The green path represents the training phase and the blue path shows the inference process. The segmentation mask is computed as the residual distance between the prediction and ground truth (see Section 5.3) . The details of both stages are discussed in more detail in Section 5. *Note:* The brain images are adapted from Pundir, Aaditya, and Rajeev Kumar. *"Brain Tumor Classification in MRI Images Using Transfer Learning."* Machine Learning for Intelligent Multimedia Analytics. Springer, Singapore, 2021. 307-319.

One fundamental challenge with the generative approach is that such models, if not well trained, may not learn efficient image representations and fail to detect subtle lesions that are difficult to distinguish from normal brain tissue. [12]. [13] addresses this problem by pre-training the encoder network on ImageNet [14]. However, it is not clear if *transfer learning* can provide a reliable translation from models pre-trained on a non-medical image dataset [15]. Another issue is that quite often diseases consist of an accumulation of anomalies and we cannot directly identify it without considering adjacent slices [16].

It can be seen that the problem of lesion detection using machine learning remains quite complex and requires many different factors to be taken into account. It must be noticed that this project does not distinguish between different types and stages of the disease, but rather explores a more global framework for detecting and segmenting any visible abnormalities on MRI scans of the human brain.

## 3.2 Self-Supervised Learning

Self-supervised learning is a way to learn meaningful representations of data without the need for human-annotated datasets. This paradigm proposes to first train the model on an auxiliary or *'pretext'* task, which is usually defined as predicting some piece of input data, and then use learned representations to solve the downstream tasks, i.e. the tasks we are actually interested in such as object detection, segmentation, etc. [17]. The idea is that by trying to solve these tasks, the model can learn some useful representation of the data relevant to the main problem. This is also more biologically plausible than the supervised approach, as it is more similar to how human make hypotheses relying on previously acquired background knowledge of how the world works [18].

Some common 'pretext' tasks are: context inpainting (filling in missing regions of the original image), relative

position (sampling two patches and predicting their position relative to each other) and colorization (predicting pixel intensity for a small patch of a grayscale image). This can be advanced by combining several self-supervision tasks in different combinations [19] and finding the combination that results in the most meaningful representation of the input image. Sometimes such models can be improved with *fine-tuning* on available small amounts of labelled data [20].

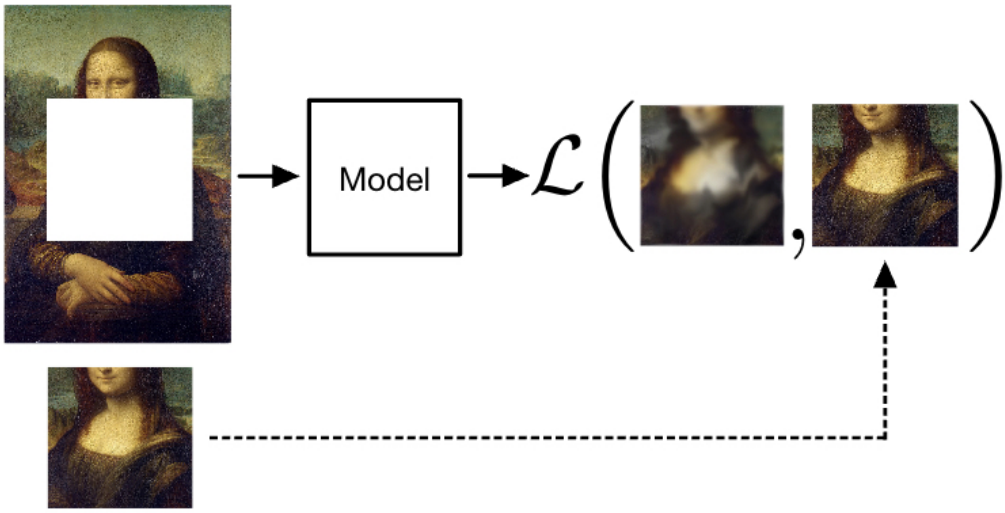


Figure 2: The visualization of the context inpainting task. Part of the image is masked and the model is trained to predict the missing part by minimizing the difference between the predicted and ground truth patches.

An approach called Context Encoders was proposed in [21] for filling in missing areas of an image. This method combines encoder-decoder architecture of AEs with adversarial loss used in GANs. The model is trained to reconstruct a missing part of an image while learning expressive latent representation. It can then be fine-tuned for various tasks, including segmentation. In [2] this method is combined with VAE and used for UAD problem. As reported in this study, this approach results in semantically richer representations and more interpretable reconstruction errors, and therefore may be useful for detecting anomalies in data.

Whether AE and VAE should be considered self-supervised models is still a matter of controversy. The term 'unsupervised' can often be misleading as it suggests that no labelling is used at all during the learning phase [18], however, autoencoders are not unsupervised in this way because they use part of the input signal as labels. In this study we assume that since these methods generate their own labels from the training data they can be called self-supervised.

# 4 Dataset

This project will use T1-weighted images from the IXI Dataset[1] to model the healthy distribution of the data in the latent space. Then, BraTS2020[2] dataset [22–24] which contains pre-operative scans of glioblastoma and low grade glioma will be used for anomaly segmentation task. Additionally, It will use ATLAS[3] dataset,

[1] https://brain-development.org/ixi-dataset/
[2] https://www.med.upenn.edu/cbica/brats2020/data.html
[3] https://fcon_1000.projects.nitrc.org/indi/retro/atlas.html.

which contains images of stroke patients, to test the model's ability to generalize to different pathology. Both 'unhealthy' datasets will use T1-weighted images, similar to the data used to train the model.

Table 1 shows how the data will be split in order to form training, validation and testing sets. However, these numbers only show the amount of images used and will change after slicing the dataset and removing irrelevant slices (described in Section 5.1).

| Dataset | Training | Validation | Testing |
|---|---|---|---|
| $\mathcal{D}_{\text{IXI}}$ | 662 | 0 | 0 |
| $\mathcal{D}_{\text{BRATS}}$ | 0 | 184 | 185 |
| $\mathcal{D}_{\text{ATLAS}}$ | 0 | 327 | 328 |

Table 1: Distribution of images between training, validation and testing datasets (before rejecting irrelevant slices).

# 5 Methods

This section gives an overview of the methodology used in my project and discusses the main architectural decisions, as well as methods for pre- and post-processing of data.

## 5.1 Pre-processing

Training volumetric models remains a significant challenge due to the large number of parameters that need to be trained and the difficulty of accommodating these models in GPU memory [11]. For this reason, high dimensional volumes will be split into 2D slices and then some of the slices will be rejected based on the size of the relevant amount of brain matter present. Another benefit of using 2D slicing is that the number of training examples will increase significantly, which can be helpful for reducing model variance. Both training and testing will be done using axial slices only. Figure 3 shows axial slices taken from approximately the middle of five volumes randomly selected from the dataset and their segmentation masks.

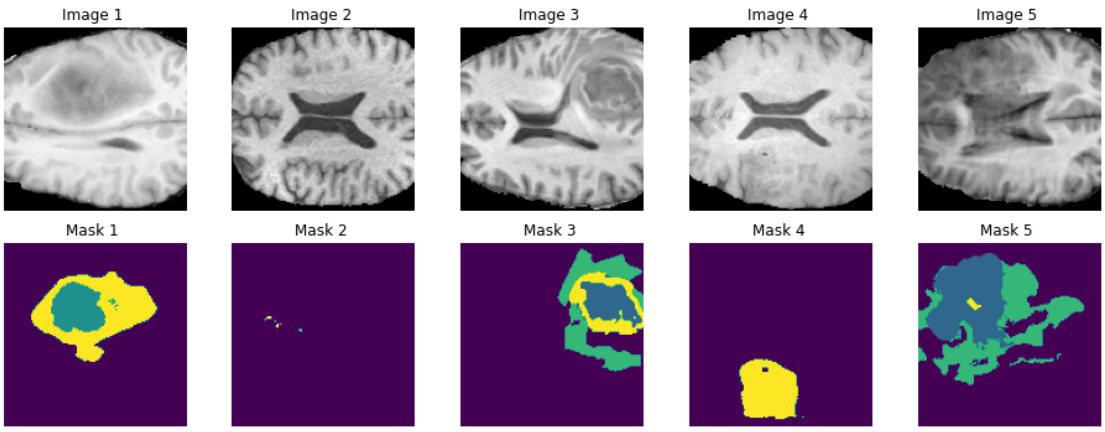


Figure 3: 2D slices from BraTS dataset and their segmentations. Different colours correspond to different segments of a tumor (yellow: the GD-enhancing tumor, turquoise: the peritumoral edema, blue: the non-enhancing tumor core).

Before we run the data through the model, it needs to be normalized and aligned to make sure all images have the same pixel intensity range and orientation. Images will also be rescaled to $128 \times 128$ resolution to improve computational efficiency of the network. Then the skull-stripping will be performed using DeepBrain[4] tool to extract brain tissue from the scans. Figure 4 illustrates the pre-processing pipeline for a single 2D slice from the dataset.

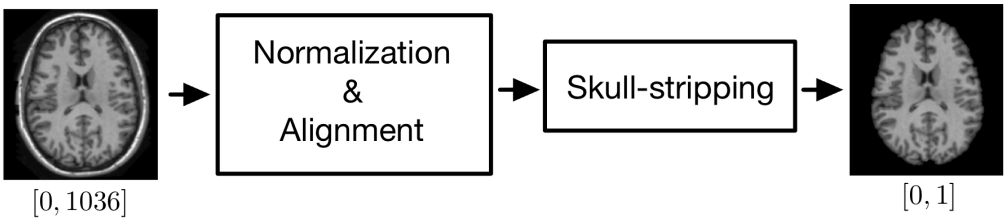


Figure 4: A pre-processing pipeline for a single 2D slice and corresponding pixel intensity ranges before and after running the example through the pipeline. *Note:* The brain image is adapted from Wang, Jiahui, et al. *"Multi-atlas segmentation of subcortical brain structures via the AutoSeg software pipeline."* Frontiers in neuroinformatics 8 (2014): 7.

## 5.2 Model Architecture

I will take the network architecture proposed in [4] as a unified baseline for AE and VAE models. The reason for using a deep neural network is because it is capable of learning a complex *non-linear* function that projects data onto a lower-dimensional manifold. As demonstrated in [25] deep autoencoders yield better data compression compared to shallow ones. Hyperparameters such as learning rate $\alpha$ and dropout rate $\delta$ will be adapted from the same study and the Adam optimizer will be used with default values of $\beta_1$ and $\beta_2$.

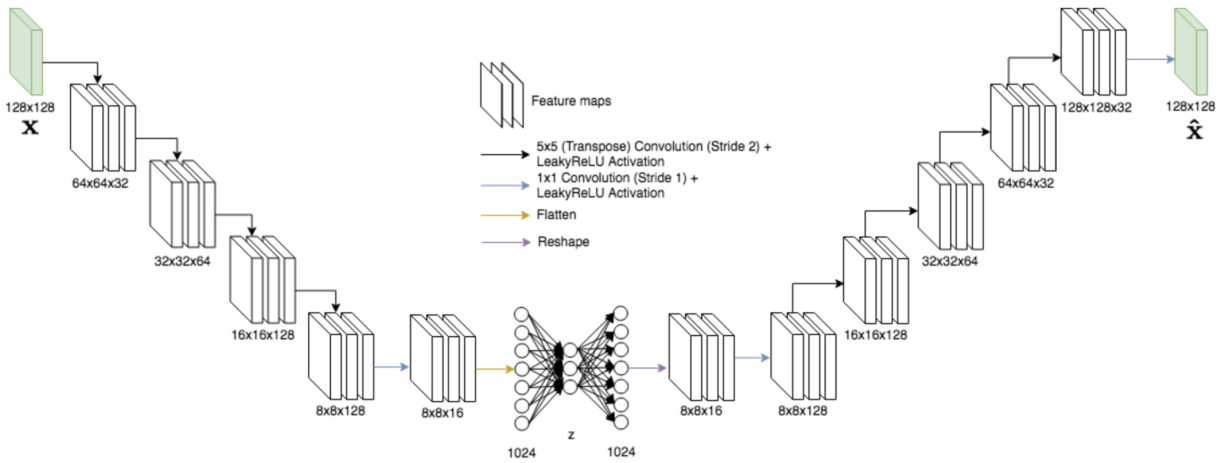


Figure 5: The unified architecture for the models used in this project. *Note:* Adapted from Baur, Christoph, et al. *"Autoencoders for unsupervised anomaly segmentation in brain MR images: a comparative study."* Medical Image Analysis 69 (2021): 101952.2

However, the shape of the bottleneck will be adjusted as a hyperparameter and several options will be

[4] `https://github.com/iitzco/deepbrain`

considered, since one of the goals of this study is to explore different latent space sizes. In the case of a dense bottleneck, it will be represented as a single vector, and in the case of a spatial one, it and flatten layers will be replaced by convolutional filters.

### 5.2.1 Autoencoder

Autoencoders (AEs) are neural network based models that attempt to learn the low dimensional representation of the original data in an unsupervised manner. Given the original image $x \in X$, AE passes it through an encoder network $Enc_\phi$ to compress it and then tries to reconstruct it as $\hat{x}$ using a decoder network $Dec_\theta$. The loss can be represented as the Mean Squared Error (MSE) between the original image and its reconstruction.

$$\mathcal{L}(x, \hat{x}; \phi, \theta) = (x - \hat{x})^2 \quad \text{where } \hat{x} = Dec_\theta(Enc_\phi(x)) \tag{1}$$

The parameters of the network are then learned to minimize the loss. Autoencoder can be considered as generalization of Principal Component Analysis (PCA), where instead of learning a simple linear projection of the data, it learns to map it non-linearly to a latent code. [26].

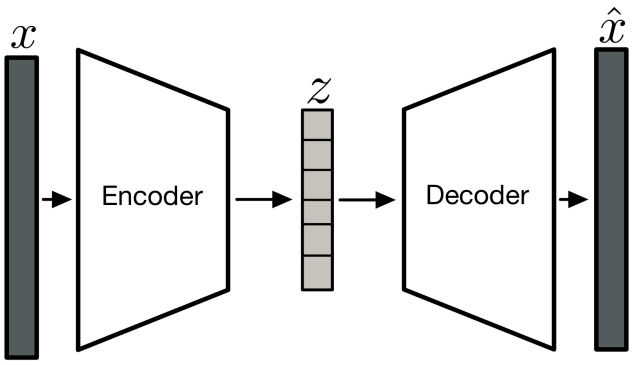


Figure 6: Basic autoencoder structure. $x$ is the original image, $z$ is the latent space, $\hat{x}$ is the reconstruction. Loss function is given in Equation 1.

### 5.2.2 Variational Autoencoder

VAE is a modified version of the autoencoder that can be used to generate new data by sampling from the latent space. The main difference from basic AE is that instead of projecting the original data onto a vector, it describes it as a probability distribution. The reason for this is that we want our latent space to group semantically similar points next to each other while keeping dissimilar points far apart. This is a problem for basic autoencoders because they often end up memorizing individual data samples by placing them in distant locations of the latent space. VAEs overcome this problem by enforcing a probabilistic prior on the latent space which penalizes the model if it tries to learn sparse representations of specific data points.

The posterior of the data given the distribution in the latent space can be expressed using Bayes Theorem:

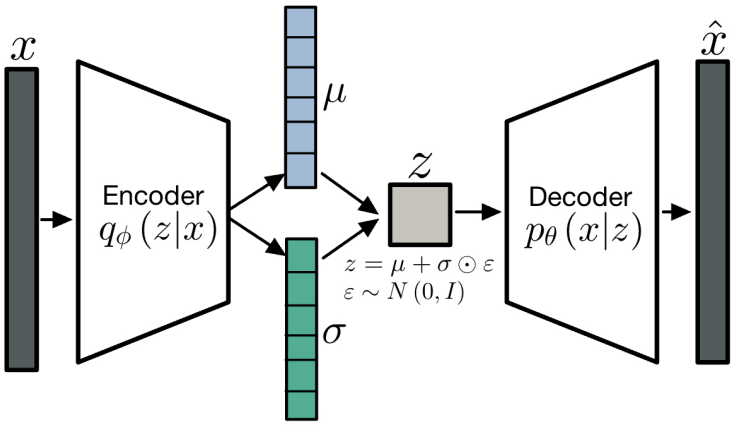


Figure 7: 'Vanilla' VAE structure. The encoder outputs two vectors for the mean and variance of the latent space, respectively. $z$ is sampled using reparametrization trick (see Equation 6), and the decoder is used to reconstruct input $x$ given the latent code $z$ as $\hat{x}$. The derivation of the loss function is discussed in Section 5.2.2 .

$$p(x|z) = \frac{p(z|x)p(z)}{p(x)} \tag{2}$$

However, the $p(x)$ term of this equation is actually an intractable integral which means that it has no closed-form solution. VAE uses variational inference to transform this complex computation problem into an optimization problem. The idea is to approximate the posterior probability distribution $p$ of the data using another distribution $q$ and find the setting of the parameters that makes $q$ close to the posterior of interest. In other words, instead of directly computing the unknown posterior probability distribution $p(z|x)$ for some data point $x \in X$, we approximate it using another distribution $q_\phi(z|x)$ which is learned via parameter optimization.

The goal of VAE is to minimize the reverse KL divergence between the ground truth posterior and approximation of the posterior:

$$D_{KL} = \mathbb{E}_{z \sim q_\phi(z|x)} \left[ \log \frac{q_\phi(z|x)}{p(z|x)} \right] \tag{3}$$

Which, after some simplifications, can be reduced to a computable loss function:

$$\mathcal{L}(x; \phi, \theta) = \underbrace{-\mathbb{E}_{z \sim q_\phi(z|x)} \Big[ \log p_\theta(x|z) \Big]}_{\text{Reconstruction Loss}} + \underbrace{D_{KL} \Big( q_\phi(z|x) \| \, p(z) \Big)}_{\text{Regularization term}} \tag{4}$$

The reconstruction loss can often be interpreted as MSE and the prior in the regularization term follows a

standard Gaussian distribution. The goal can then be represented as follows:

$$\phi^*, \theta^* = \arg\min_{\phi,\theta} \sum_{i=1}^{N} \mathcal{L}(x^{(i)}; \phi, \theta) \tag{5}$$

This is equivalent to maximizing the *variational lower bound* on the log-likelihood of the data.

Both the encoder $q_\phi(z|x)$ and the decoder $p_\theta(x|z)$ are trained synchronously using backpropagation. However, the sampling of $z$ is a stochastic process, which makes the entire network non-differentiable. To overcome this, VAE exploits a reparametrization trick that allows to divert a non-differentiable sampling operation out of the network:

$$z = \mu(x) + \sigma(x) \odot \epsilon \quad \text{where } \epsilon \sim \mathbb{N}(0, 1) \tag{6}$$

#### 5.2.3 Extensions

This project will also explore several variations of VAE that can possibly produce better results compared to basic models.

1. Vector Quantized-Variational Autoencoder (VQ-VAE). The main difference from "vanilla" VAE is that instead of learning a continuous distribution of data in the latent space, VQ-VAE learns a discrete one, and the prior is learned rather than predefined. It uses a codebook, represented as a list of vectors that are also learned, to quantize the bottleneck of the encoder's network, and then one of these vectors is passed to the decoder for reconstruction.

2. Gaussian Mixture Variational Autoencoder (GMVAE). In this architecture, the VAE unimodal prior is replaced by a Gaussian mixture. According to [7], such extension may present a more naturally suited approach and allow to fit a more complex distribution to the samples in the latent space. Although this method was originally developed for clustering task, it has been applied to the problem of lesion detection in [27].

These architectures will be discussed in more detail in the final version of this project.

### 5.3 Lesion Segmentation

Once the model is trained, we use the data from the healthy dataset to evaluate how well our model is able to reconstruct a healthy image, and if the results are plausible, we can move on to the problem of lesion detection. Since the model is trained to only reconstruct healthy images, if an abnormal image is passed during inference, it will not be able to reconstruct it properly, which should result in a blurred area around the lesion. Taking the difference between the input image and its reconstruction, we get a residual mask, which can then be binarized and corrected using morphological transformations.

# 6 Evaluation Metrics

This section discusses the metrics used to evaluate and compare different models.

## 6.1 Intersection-over-Union (IOU)

$IoU$ allows to measure the overlap between two segmentation masks $x$ and $\hat{x}$. It compares the intersection size of the two masks with the union of both predicted and ground truth segmentations.

$$IoU = \frac{x \cap \hat{x}}{x \cup \hat{x}} = \frac{TP}{TP + FP + FN} \tag{7}$$

## 6.2 Dice Score

The Dice coefficient is defined similarly to $IoU$, but instead of computing the union, we sum the areas of the two masks. This is equivalent to taking the harmonic mean or calculating the $F1$ score. This metric is more resistant to class imbalance and more likely to choose the smallest score between precision and recall, meaning that if we maximize it, it will try to balance between the two.

$$DICE = \frac{x \cap \hat{x}}{|x| + |\hat{x}|} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{8}$$

## 6.3 Area Under the Receiver-Operating Characteristic Curve (AUROC)

AUROC is area under the curve where $x$ axis is the False Positive Rate (FPR) and the $y$ axis represents the True Positive Rate (TPR). This method allows you to compare different models with different hyperparameter values. By calculating the area under each curve, we get an estimate of how good a particular method is compared to other methods.

## 6.4 Area Under the Precision-Recall Curve (AUPRC)

FPR may not always be indicative of segmentation accuracy, for example, if there are relatively few false positives compared to true negatives. AUPRC calculates the area under the curve, where the $y$-axis is precision and the $x$-axis is recall, which can be more useful when dealing with imbalanced classes.

# 7 Professional Considerations

This section addresses professional and ethical issues that may be relevant to this research project.

## 7.1 BCS Code of Conduct

All work on this project will be carried out in accordance with the BCS Code of Conduct [5]. The relevant sections of this document are as follows:

1. **Public Interest**

   Personal data coming from third parties will be protected and carefully used. All intellectual property of other authors will be properly interpreted and referenced.

2. **Professional Competence and Integrity**

   The author can claim to have some experience in this research area having studied the Fundamentals of Machine Learning and Computer Vision modules at the University of Sussex. The project will follow the best coding and academic writing practices related to the authors' area of specialism. Working on this project will help the author to expand knowledge in the field of machine learning and will serve as a basis for future research work.

3. **Duty to Relevant Authority**

   The author accepts full professional responsibility for the work done in this project and claims that all work was done by the author himself and does not contain examples of plagiarism, collusion or fabrication of results. The results of the study will be presented in a clear and concise manner without withholding any information about system performance. If any conflict arises during this study, it will be reported to the university immediately.

4. **Duty to the Profession**

   Where necessary, support for other community members will be provided through contributing to open source projects relevant to the area of this research. This study will uphold the reputation of BCS, the Chartered Institute of Information Technology, and the computer scientist profession.

## 7.2 Ethical Review

Since human data is used in this research project, it is important to consider relevant ethical aspects. The data used for this study was pre-collected by medical institutions and does not contain any information that could identify participants (e.g. race, religion, sexual orientation, biometrics, etc.). During the development of this research project, it will be processed and analyzed using various machine learning methods. Each dataset contains a CSV file with some demographic information such as gender, age, and date of study, which may be useful for working on the domain shift problem, but does not contain detailed information about patients' personal data. In addition, some of the datasets provide a unique identifier associated with each patient. Nevertheless, none of this information can be used to identify a patient or attempt to contact them directly.

All research data will be stored securely on researchers' OneDrive and will not be shared with third parties including other students. The data will only be used for the purposes of this study and will not be stored for

[5] `https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/`

use in any other projects. All work on these data will be carried out only in the laboratories of the University of Sussex. The ethical review application is attached to this report as an Appendix B.

# 8 Requirements Analysis

This section describes the requirements the system must meet and the specific properties it must have.

## 8.1 Functional Requirements

- The system can be used by medical experts as a tool to perform lesion segmentation tasks.
- The system is capable of performing anomaly segmentation tasks without relying on supervised learning methods.
- The project implements several types of anomaly detection methods and allows the user to select the preferred one.
- The system can handle pre-processing, inference, and post-processing of the new data if this data is presented in the correct format, i.e. as a NiBabel or NumPy array.
- The predicted segmentation masks are in the form of a binary image and can subsequently be used by trained physicians.
- The system is resistant to domain shifts such as other diseases, age groups, and scanner types.
- The system is relatively accurate compared to existing lesion segmentation methods, and if not, the techniques for improving its accuracy are discussed.

## 8.2 Non-functional Requirements

- During inference, the system runs in a fairly short time, i.e. the prediction is made in less than 2 minutes.
- The project code and documentation are clear and written according to best practices.
- The project ensures that all personal data is stored securely.
- The project is reliable and will work without failures under predetermined conditions (e.g. if run on a modern computer with the minimum required characteristics).
- Package versions used in the system do not conflict with each other.
- The project can be imported from the GitHub repository and run on any machine with the appropriate technical specifications.

# 9 Project Plan

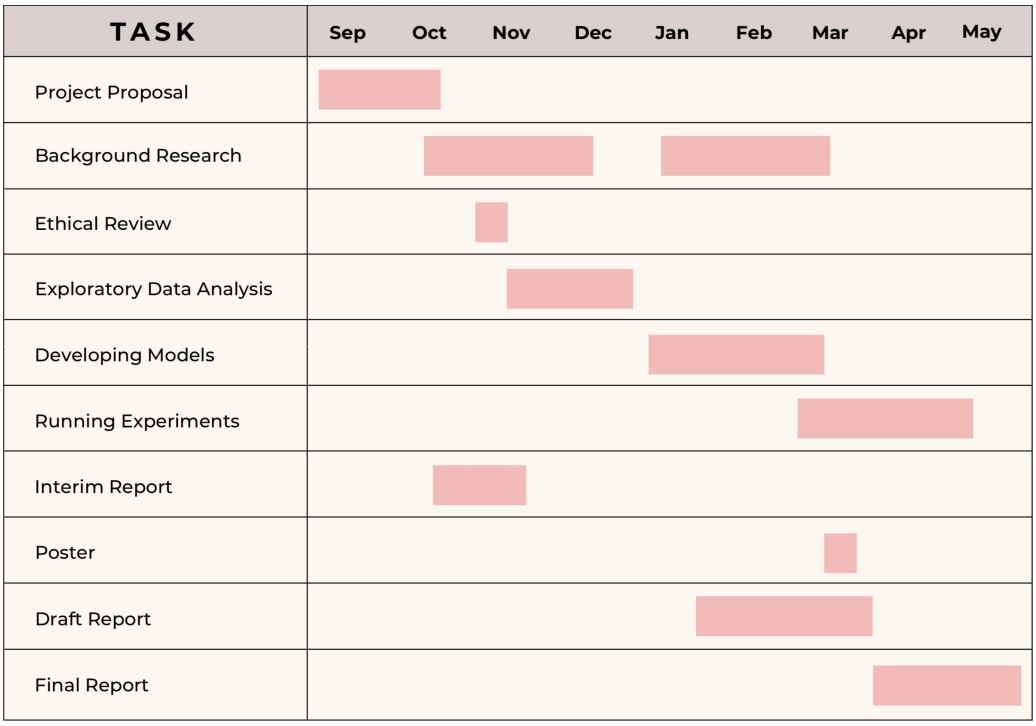


Figure 8: Gantt chart representing project timescale.

# 10 Summary

This project explores the field of anomaly detection in MRI scans of the human brain, with a focus on self-supervised learning methods. The main goal is to study some of these methods and explore how well they can be generalized to different data domains. In order to do that this project will use several datasets coming from different sources to first learn the healthy data distribution and then use the trained model to detect abnormal areas in unhealthy images. After that, the models will be evaluated and the results presented. In addition, this project will look at the effect of using different latent space sizes and examine if the size of the training data used is sufficient to obtain accurate segmentations. Finally, possible directions for future research will be discussed.

# References


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# A Meeting Log

- **Meeting 1, 03/10/2022**
  - Discussed several self-supervised learning methods and potential directions for my project.
  - Discussed some of the relevant datasets that can be used for model training and inference.
  - Discussed packages and frameworks that can be useful for the project.
  - Agreed on time for weekly group meetings.
- **Meeting 2, 14/10/2022**
  - Each of the group members made a short presentation, revealing the main ideas of their project.
  - Discussed that there are two ways to perform lesion segmentation: either reconstruction of the original image or latent space analysis.
  - Discussed that domain shift can be studied not only in relation to other diseases, but also in relation to other age groups or types of scanners.
- **Meeting 3, 21/10/2022**
  - Discussed several methods for removing the skull on MRI images of the brain.
  - Discussed the details of ethical application and how the demographic data coming with the datasets can be used.
  - Discussed a method for plotting the distribution of data without including the background pixels.
  - Briefly discussed VQ-VAE and GMVAE, along with related literature that may be helpful in understanding these architectures.
- **Meeting 4, 04/11/2022**
  - Discussed expectations from the interim report (Functional/non-functional requirements, Gantt Chart, Relevant work).
  - Discussed Conditional Variational Autoencoders and the potential of including them into the project.
  - Discussed that diagrams and equations in a dissertation should be labeled and referenced appropriately.
  - Answered questions from other members of the group and shared some tips for using LaTeX.

# B Ethical Review Application

## Ethical Review Application (ER/SK768/1) Sergey Kuznetsov

| | |
|---|---|
| **Project Title** | Self-Supervised Machine Learning On MRI Data |
| **Status** | With supervisor |
| **Email** | sk768@sussex.ac.uk |
| **Phone No.** | 07749786250 |
| **Applicant Status** | UG |
| **Department** | Informatics |
| **Supervisor** | Simpson, Ivor JA |
| **Project Start Date** | 01-Nov-2022 |
| **Project End Date** | 31-May-2023 |
| **External Funding in place** | No |
| **External Collaborators** | No |
| **Funder/Project Title** | |
| **Name of Funder** | |
| **Project Description** | My project will explore some of the available self-supervised machine learning techniques for lesion detection and segmentation on Magnetic Resonance Imaging (MRI) scans. Lesion segmentation plays a critical role in a variety of clinical tasks such as medical diagnosis, prognosis, and treatment. In most cases, it is performed manually by the radiologist, making the process very time-consuming and dependent on understanding of the human anatomy and expertise. Recent advances in deep learning, and generative modelling in particular, have proven to be promising tools for detecting arbitrary anomalies in data and automating medical analysis, relieving professionals from time-consuming tasks. However, training supervised models can often be a significant challenge. The reason for this is that training neural networks typically requires large amounts of labelled data, which is often difficult to obtain due to disease heterogeneity and patient privacy aspects. Therefore, we need to consider such machine learning models that do not require vast amounts of labelled data and can properly generalize to rare diseases that may not have been seen during training. Self-supervised methods require smaller amounts of labelled data needed to properly train the model, and sometimes may not require any labelling at all. My project will use multiple MRI datasets coming from different sources, one of them will be used to model the pixel intensity distribution of a healthy brain and the other two will be used to segment lesions. All of them contain MRI data of human patients, and two of them also contain corresponding lesion segmentation masks. In addition, each dataset also contains a CSV file with some demographic information about patients (weight, sex, age, study date, etc.). |

| Ethical Review Form Section A (ER/SK768/1) | |
|---|---|
| **Question** | **Response** |
| | |
| >> Checklist | |
| A1. Will your study involve participants who are currently or potentially vulnerable or unable to give informed consent or in a dependent position (e.g. people under 18, people with learning difficulties, over-researched groups or people in care facilities)? | No |
| A2. Will participants be required to take part in the study without their consent or knowledge at the time (e.g. covert observation of people in non-public places), and / or will deception of any sort be used? Please refer to the British Psychological Society Code of Ethics and Conduct (or similar guidelines) for further information. | No |
| A3. Unless specifically and clearly consented (e.g. a media release form), will it be possible, through a research output, to identify participants in any way? (This does not include taking email details for participant prize draws or identifying participants from signed consent forms or holding identity encryption spreadsheets that are stored securely separate from the research data). | No |
| A4. Might the study induce psychological stress or anxiety, or produce humiliation or cause harm or negative consequences beyond the risks likely to be encountered in the everyday life of the participants? | No |
| A5. Is there a risk that the research topic might lead to disclosures from the participant concerning their beliefs, involvement in illegal actions or any other activities that may represent a threat to themselves or others? | No |
| A6. Will the study involve collecting any personal special category information* in a form that could allow the participant/ participants to be identified?<br>[* identifiers relating to race, ethnic origin, politics, religion, trade union membership, philosophical beliefs, genetics, biometrics, health, sex life or sexual orientation] | No |
| A7. Will any drugs, placebos or other substances (such as food substances or vitamins) be administered as part of this study and will any invasive or potentially harmful procedures of any kind will be used? | No |
| A8. Will your project involve working with any substances and / or equipment which may be considered hazardous? | No |
| A9. Will your study involve the taking and/or storage of human tissue that falls under the Human Tissue Act (HTA)?<br>http://www.sussex.ac.uk/staff/research/governance/erp_overview/humantissue | No |
| >> Risk Assessment | |
| A10. If you have answered Yes to ANY of the above questions, your application may be considered as HIGH risk. If, however you wish to make a case that your application should be considered as LOW risk please enter the reasons here. Researchers should note that SREOs or C-RECs may decide NOT to agree with the case that you have made. | |

| Ethical Review Form Section B (ER/SK768/1) | |
|---|---|
| **Question** | **Response** |
| | |
| >> Data Collection and Analysis (Please provide full details) | |
| B1. PARTICIPANTS: How many people do you envisage will participate, who are they, and how will they be selected? | First dataset: 600 subjects.<br>Second dataset: approximately 350 subjects.<br>Third dataset: approximately 650 subjects (Note: it is possible that only a limited amount of data from this dataset will be used, as the purpose of using it is to study the robustness of the machine learning model to domain shift).<br>Total: approximately 1600 data samples. The data was acquired with various scanners in different hospitals and was manually segmented by radiologists.<br>The identity of all subjects is unknown and we can only see their demographic details such as age, sex and survival information. |
| B2. RECRUITMENT: How will participants be approached and recruited? | The data has been pre-collected, so no one will try to contact the participants directly. |
| B3. METHOD: What research method(s) do you plan to use; e.g. interview, questionnaire/self-completion questionnaire, field observation, audio/audio-visual recording? | The data will be processed and analyzed using various machine learning methods. |
| B4. LOCATION: Where will the project be carried out e.g. public place, in researcher's office, in private office at organisation? | University of Sussex labs. |
| B5. PARTICIPANT WELLBEING: Will the study involve engaging participants in the discussion of potentially distressing or sensitive topics? (e.g. sexual activity, drug use, ethnicity, political behaviour, potentially illegal activities). If so, please set out how you will manage the well-being of participants. | No. |
| >> Confidentiality and Anonymity | |
| B6. Will questionnaires be completed anonymously and returned indirectly? | N/A |
| B7. Will research data only be identifiable by a unique identifier (e.g. code/pseudonym)? If Yes, please explain how this will be attributed in B11a below. | N/A |
| B8. Will lists of identity numbers or pseudonyms linked to names and/or addresses be stored securely and separately from the research data? If Yes, explain how this will occur in B11a below. | N/A |
| B9. Will all place names and institutions which could lead to the identification of individuals or organisations be changed unless this is consented to explicitly in the consent form? | N/A |
| B10. Will all personal information gathered be treated in strict confidence and never disclosed to any third parties? | N/A |
| B11. Can you confirm that your research records will be held in accordance with data protection regulations? (http://www.sussex.ac.uk/ogs/policies/information/dpa) | Yes |

| | |
|---|---|
| B11a. Please explain how ANY identifiable personal and/or research data will be managed and securely stored ensuring that participants have given appropriate informed consent for this. | Although the data is not identifiable, it will be securely stored on the researcher's OneDrive space. |
| B12. Do you intend to use the research data for any purpose other than that for which consent is explicitly given? If so, please explain below | No |
| B12a. If you answered NO to any of the above in this section (or think more information could be useful to the reviewer) please explain here: | |
| >> Informed Consent and Recruitment of Participants | |
| B13. Will all respondents be given an Information Sheet and be given adequate time to read it before being asked to agree to participate? | N/A |
| B14. Will all participants taking part in an interview, focus group, observation (or other activity which is not questionnaire based) be asked to sign a consent form? If you are obtaining consent another way (such as verbally), please explain under B17 below. | N/A |
| B15. Will all participants self-completing a questionnaire be asked to show consent to participate by a specific and identifiable action? (Give details in B17 below) | N/A |
| B16. Will all participants be told that they can withdraw their participation at any time during the research and can ask for their data to be destroyed and/or removed from the project until it is no longer practical to do so? | N/A |
| B17. If you answered NO to any of the above in this section (or think more information will be useful to the reviewer) please explain here: | |
| >> Context | |
| B18. Is DBS (Disclosure and Barring Service) clearance necessary for this project? If yes, please ensure you complete the next question. | No |
| B19. Are any other ethical clearances or permissions (internal or external) required? Please see the help text (i) for further details. | No |
| B19a. If yes, please give further details including the name and address of the organisation. If other ethical approval has already been received please attach evidence of approval, otherwise you will need to supply it when ready. (You do not need to provide evidence of a current DBS check at this point). | |
| B20. Does the research involve any fieldwork - Overseas or in the UK? | No |
| B20a. If yes, where will the fieldwork take place? If undertaken overseas you must attach an OTSSRA form. In the event that the Foreign and Commonwealth Office has specific travel warnings in place for the country (ies) to be visited you will also need to provide a detailed risk assessment. https://www.gov.uk/foreign-travel-advice | |

| | |
|---|---|
| B21. Will any researchers be in a lone working situation? | No |
| B21a. If yes, briefly describe the location, time of day and duration of the lone working. What precautionary measures will be taken to ensure safety of the researcher(s)? | |
| >> Any further concerns | |
| B22. Are there any other ethical considerations relating to your project which have not been covered above? | No |
| B22a. If yes, please explain: | |

# C Project Proposal

# Self-Supervised Learning on MRI Data

Sergey Kuznetsov

October 2022

## 1 Aims

Lesion segmentation plays a critical role in a variety clinical tasks such as medical diagnosis, prognosis, and treatment. In most cases, segmentation is performed manually by the radiologist, making the process very time-consuming and dependent on understanding of the human anatomy and expertise. Recent advances in deep learning, and generative modeling in particular, have proven to be promising tools for detecting arbitrary anomalies in data and automating medical analysis, relieving professionals from time-consuming tasks [1]. However, training supervised models can often be a significant challenge. The reason for this is that training neural networks typically requires large amounts of labeled data, which is often difficult to obtain due to disease heterogeneity and patient privacy aspects. Therefore, we need to consider such machine learning models that do not require vast amounts of labeled data and can properly generalize to rare diseases that may not have been seen during training. One of the possible approaches is to use models based on variational inference to model the healthy distribution of the human brain in a low-dimensional space and then, during inference, detect anomalous data passed as input to that model as outlier. This approach is more similar to how radiologists make their decisions, as it does not require manual annotation and relies only on a priori knowledge of pathologies manifestations. [2]. Although there are several possible ways to address this task, our project will focus mainly on probabilistic graphical models such as Variational Autoencoder (VAE) [3] and it's extensions [4] [5]. VAEs learn the distribution of the data by mapping it to the normative distribution in the latent space and do not require any labels for learning. Once trained, the model will produce different latent encodings for the unhealthy samples as they will have a different distribution than the one it has learned. Their reconstructions can then be segmented via various morphological operators or MAP-based methods [6] and compared to the ground truth segmentations.

## 2 Objectives

### 2.1 Primary Objectives

1. Investigate and compare some of the available medical image analysis methods for lesion detection and segmentation.
2. Implement the described architectures in Python using PyTorch deep learning framework.

3. Examine the performance of Autoencoders, Variational Autoencoders in relation to lesion detection and segmentation tasks.
4. Research the effects of domain shift and the ability of models to generalize to new data with different distributions.

### 2.2 Extension Objectives

1. Explore the extensions of VAEs and compare their results to classic VAE and AE results.
2. Study the weaknesses of the demonstrated methods and discuss potential improvements.
3. Examine the amount of data needed for proper model generalization and inference.
4. Study how the size of the latent space affects the quality of segmentation.

## 3 Dataset

Our project will use T1-weighted images from the IXI Dataset[1] to model the healthy distribution of the data in the latent space. Then, BraTS2020 dataset [7–9] which contains pre-operative scans of glioblastoma and lower grade glioma will be used for anomaly detection and segmentation tasks. In addition, we will use ATLAS[2] dataset to test the ability of the model to generalize to new domains. Both 'unhealthy' datasets will use T-1 weighted images, similar to the data used to train the model.

## 4 Relevance

Generative modeling has already shown impressive results in anomaly detection, but machines are still not as good at this task as humans. Since the number of experts who can detect the anomaly does not grow as far as the number of patients to be studied [10], we need to explore new approaches to automate this process. Our task is to explore some of the state-of-the-art methods and evaluate how good they are for a lesion segmentation task. As mentioned, this approach is more human-like than supervised deep learning, and, in a broader perspective, could give us a more reliable and faster way to detect and treat certain diseases at early stages.

[1] https://brain-development.org/ixi-dataset/
[2] https://fcon_1000.projects.nitrc.org/indi/retro/atlas.html.

# 5 Project Timescale

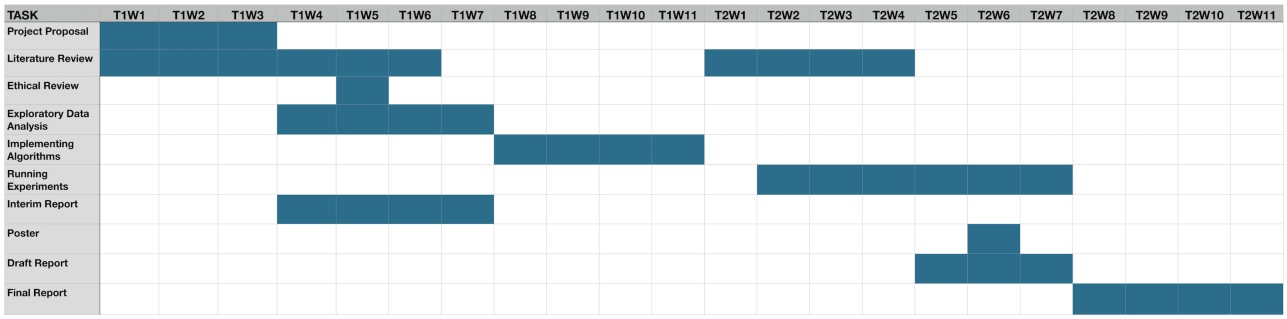


Figure 1: Project timescale.

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